

ITA0605 - Machine Learning

List of Lab Exercises

19. To predict bank loan approval using the Naïve Bayes algorithm.

Aim:

To predict bank loan approval using the Naïve Bayes classification algorithm.

Algorithm:

- 1.Import the required Python libraries and Naïve Bayes classifier.
- 2.Create the bank loan dataset with input features and target labels.
- 3.Separate the input features (X) and target variable (y).
- 4.Train the Naïve Bayes model using the dataset.
- 5.Provide the new applicant details as input.
- 6.Predict whether the loan is Approved or Rejected.
- 7.Display the predicted loan approval result.

Dataset:

Program Code:

```
import pandas as pd

from sklearn.naive_bayes import GaussianNB

data = {

    'Income': [12,10,9,7,6,5,4,8,3,11],

    'CreditScore': [820,790,760,700,680,650,620,730,600,810],

    'LoanAmount': [5,6,7,8,9,10,12,8,13,5],

    'LoanStatus': ['Approved','Approved','Approved','Approved','Rejected',

                   'Rejected','Rejected','Approved','Rejected','Approved']

}
```

```
df = pd.DataFrame(data)

X = df[['Income', 'CreditScore', 'LoanAmount']]

y = df['LoanStatus']

model = GaussianNB()

model.fit(X, y)

new_sample = [[9, 770, 6]]

prediction = model.predict(new_sample)

print("Predicted Loan Status:", prediction[0])
```

Output:

```
# New sample
new_sample = [[9, 770, 6]]

# Prediction
prediction = model.predict(new_sample)

print("Predicted Loan Status:", prediction[0])

... Predicted Loan Status: Approved
```

Dataset 2:

Program Code:

```
import pandas as pd

from sklearn.naive_bayes import GaussianNB

data = {

    'Salary': [90000,80000,70000,60000,50000,45000,40000,75000,35000,95000],

    'CIBIL': [850,820,780,740,690,670,640,800,620,870],

    'EMI': [8000,9000,12000,15000,18000,20000,23000,10000,25000,7000],

    'LoanStatus': ['Approved','Approved','Approved','Approved','Rejected',

                   'Rejected','Rejected','Approved','Rejected','Approved']
```

```
}
```

```
df = pd.DataFrame(data)
```

```
X = df[['Salary', 'CIBIL', 'EMI']]
```

```
y = df['LoanStatus']
```

```
model = GaussianNB()
```

```
model.fit(X, y)
```

```
new_sample = [[72000, 790, 11000]]
```

```
prediction = model.predict(new_sample)
```

```
print("Predicted Loan Status:", prediction[0])
```

Output:

```
model.fit(X, y)

# New sample
new_sample = [[72000, 790, 11000]]

# Prediction
prediction = model.predict(new_sample)

print("Predicted Loan Status:", prediction[0])

... Predicted Loan Status: Approved
```

Dataset 3:

Program Code:

```
import pandas as pd
```

```
from sklearn.naive_bayes import GaussianNB
```

```
data = {
```

```
    'BusinessIncome': [20,18,15,12,10,8,6,16,5,22],
```

```
    'BusinessAge': [10,9,8,6,5,4,3,8,2,12],
```

```
    'ExistingLoans': [1,1,2,2,3,3,4,2,5,1],
```

```
'LoanStatus': ['Approved','Approved','Approved','Approved','Rejected',  
               'Rejected','Rejected','Approved','Rejected','Approved']  
}
```

```
df = pd.DataFrame(data)
```

```
X = df[['BusinessIncome', 'BusinessAge', 'ExistingLoans']]
```

```
y = df['LoanStatus']
```

```
model = GaussianNB()
```

```
model.fit(X, y)
```

```
new_sample = [[17, 8, 2]]
```

```
prediction = model.predict(new_sample)
```

```
print("Predicted Loan Status:", prediction[0])
```

Output:

```
# New sample  
new_sample = [[17, 8, 2]]  
  
# Prediction  
prediction = model.predict(new_sample)  
  
print("Predicted Loan Status:", prediction[0])
```

```
... Predicted Loan Status: Approved
```

Dataset 4:

Program Code:

```
import pandas as pd

from sklearn.naive_bayes import GaussianNB

data = {

    'FamilyIncome': [15,13,11,9,7,6,5,12,4,16],

    'AcademicScore': [92,89,85,80,72,68,65,87,60,94],

    'LoanAmount': [6,7,8,9,10,11,12,7,13,5],

    'LoanStatus': ['Approved','Approved','Approved','Approved','Rejected',

                   'Rejected','Rejected','Approved','Rejected','Approved']

}

df = pd.DataFrame(data)

X = df[['FamilyIncome', 'AcademicScore', 'LoanAmount']]

y = df['LoanStatus']

model = GaussianNB()

model.fit(X, y)

new_sample = [[12, 88, 7]]

prediction = model.predict(new_sample)

print("Predicted Loan Status:", prediction[0])
```

Output:

```
# New Sample
new_sample = [[12, 88, 7]]

# Prediction
prediction = model.predict(new_sample)

print("Predicted Loan Status:", prediction[0])

... Predicted Loan Status: Approved
```

Dataset 5:

Program Code:

```
import pandas as pd

from sklearn.naive_bayes import GaussianNB

data = {

    'Income': [14,12,10,8,7,6,5,11,4,15],

    'DownPayment': [3,2.5,2,1.8,1.5,1.2,1.0,2.2,0.8,3.5],

    'CreditScore': [830,810,780,740,690,660,630,790,600,850],

    'LoanStatus': ['Approved','Approved','Approved','Approved','Rejected',

                   'Rejected','Rejected','Approved','Rejected','Approved']

}

df = pd.DataFrame(data)

X = df[['Income', 'DownPayment', 'CreditScore']]

y = df['LoanStatus']

model = GaussianNB()

model.fit(X, y)

new_sample = [[11, 2.4, 800]]
```

```
prediction = model.predict(new_sample)

print("Predicted Loan Status:", prediction[0])
```

Output:

```
# New sample
new_sample = [[11, 2.4, 800]]

# Prediction
prediction = model.predict(new_sample)

print("Predicted Loan Status:", prediction[0])

... Predicted Loan Status: Approved
```

Result:

The Naive Bayes algorithm was successfully implemented on all five loan approval datasets. The model successfully predicted the loan status for each given new sample. Thus, bank loan approval prediction was successfully performed using Naive Bayes.